

OJAS GRAMOPADHYE

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EDUCATION

New York University, New York Aug 2024 - May 2026
Master of Science, Computer Science GPA : 3.89/4
Courses : Artificial Intelligence, Deep Learning, Robot Perception, Big Data, Algorithms, ML Sys-Ops (Spring 2026)

Indian Institute of Technology Bombay, India May 2019 - Jul 2023
Bachelor of Technology, Computer Science and Engineering GPA : 7.74/10
Courses : AI & ML, Speech Recognition, Image Processing, Database Systems, Data Structures, Software Systems

KEY PROJECTS

Flowsight : NYC Traffic Collision Analysis and Prediction Jan 2025 - May 2025
Course Project | Big Data **New York University**

- Processed millions of NYC traffic and collision records using a Kafka - MongoDB - Spark pipeline, enriching speed data with crash context through BallTree-based haversine nearest-neighbor joins
- Identified high-risk collision hotspots by training a Spark ML model, thresholding outputs into custom risk function based on incident impact, monitoring with a React dashboard and d3.js visualizations

Erica : AI Tutor agent with GraphRAG Sept 2025 - Dec 2025
Course Project | Artificial Intelligence **New York University**

- Developed a Dockerized LLM tutor agent that ingests multi-format course content, enriches it with LLM metadata, stores corpus artifacts in MongoDB, and pipelines them into GraphRAG for grounded, concept-aware retrieval
- Delivered structured, prerequisite-aware explanations by integrating semantic search, a typed knowledge graph, and Streamlit UI with subgraph visualizations, enabling the agent to answer queries with references

Multi-subject alignment : Neural Speech Decoding Sept 2025 - Ongoing
Graduate Research | NYU Video Lab **New York University**

- Improved cross-subject neural speech decoding by learning subject-invariant embeddings through contrastive alignment and adapter-based modeling across multiple ECoG participants
- Experimenting with representation learning by implementing a wav2vec2-based pretraining pipeline : masked encoder modeling, Gumbel quantization, negative sampling, and a Transformer context network for phoneme classification

Teaching Assistant roles Jun '25 - Ongoing

- Grader - Object Oriented Programming** (Prof. Hasan Aljabbouli) : Graded UML diagrams, Java/C++ assignments, and design pattern projects for 60+ students, ensuring consistent evaluation across weekly submissions in the Object-Oriented Programming course.
- Teaching Assistant - Machine Learning** (Prof. Mina Ghashami) : Supporting the Machine Learning course by designing assignments, creating grading rubrics, evaluating student implementations across core ML algorithms and neural networks, and interacting with students to clarify concepts and guide project development.

WORK EXPERIENCE

Detection and Mitigation of Hallucination in Generative Models May 2023 - May 2024
Research Assistant **IBM Research | IIT Bombay**

- Improved diagnostic accuracy of LLMs to 90% by designing a Forward-Backward Chain-of-Thought evaluation pipeline and fine-tuning LLaMA-2 (7B, 70B) models for hallucination mitigation in medical QA
- Built a Subjective Medical QA benchmark and a reward-based verification pipeline that generates mixed-step reasoning signals and enables scalable, reproducible experimentation - EMNLP Findings 2024 ([arXiv: arxiv.org/abs/2403.04890](https://arxiv.org/abs/2403.04890))

Japanese OCR Pipeline May 2022 - Jul 2022
Data Science Intern **Daikin**

- Delivered a full OCR pipeline for 3,000+ Japanese documents by building document-cleaning and image-normalization scripts, achieving 0.91 character-level accuracy with a MobileNetV3 + Differentiable Binarization detector
- Wrote preprocessing and evaluation pipeline to streamline dataset preparation and measure OCR performance on proprietary scanned documents

TECHNICAL SKILLS

Machine Learning: LLMs, RAG systems, AI agents, NLP, deep learning, contrastive learning, representation learning, LoRA fine-tuning
Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace, PyTorch Lightning, LangGraph, LangChain, SentenceTransformers, scikit-learn
Data & Systems: Apache Kafka, Apache Spark, Dask, MongoDB, PostgreSQL, ChromaDB, MLflow, Docker, Linux, Bash
Software Engineering: Python, C++, JavaScript, TypeScript, FastAPI, React, Streamlit, d3.js, Git, REST APIs